



UNIMORE
UNIVERSITÀ DEGLI STUDI DI
MODENA E REGGIO EMILIA

Dipartimento di Ingegneria
"Enzo Ferrari"

Processi di lavorazione per materiali polimerici e compositi

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Introduzione

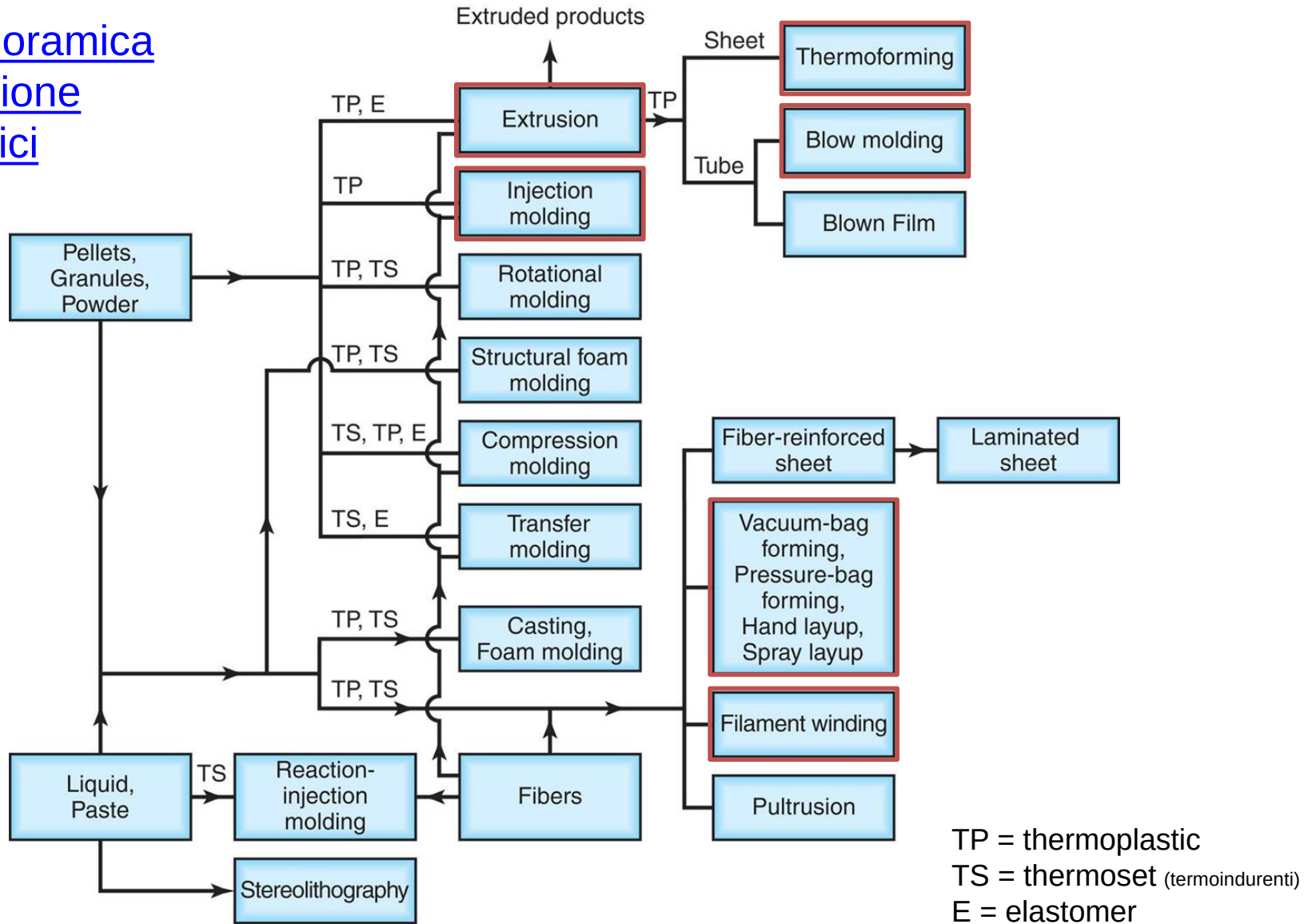
General Characteristics of Forming and Shaping Processes for Plastics and Composite Materials

Process		Characteristics
Extrusion	Polimeri termoplastici	Continuous, uniformly solid or hollow, and complex cross-sections; high production rates; relatively low tooling costs; wide tolerances
Injection molding		Complex shapes of various sizes; thin walls; very high production rates; costly tooling; good dimensional accuracy
Structural foam molding		Large parts with high stiffness-to-weight ratio; less expensive tooling than in injection molding; low production rates
Blow molding		Hollow, thin-walled parts and bottles of various sizes; high production rates; relatively low tooling costs
Rotational molding		Large, hollow items of relatively simple shape; relatively low tooling costs; relatively low production rates
Thermoforming	Termoindurenti	Shallow or relatively deep cavities; low tooling costs; medium production rates
Compression molding		Parts similar to impression-die forging; expensive tooling; medium production rates
Transfer molding		More complex parts than compression molding; higher production rates; high tooling costs; some scrap loss
Casting		Simple or intricate shapes made with rigid or flexible low-cost molds; low production rates
Processing of composite materials		Long cycle times; expensive operation; tooling costs depend on process

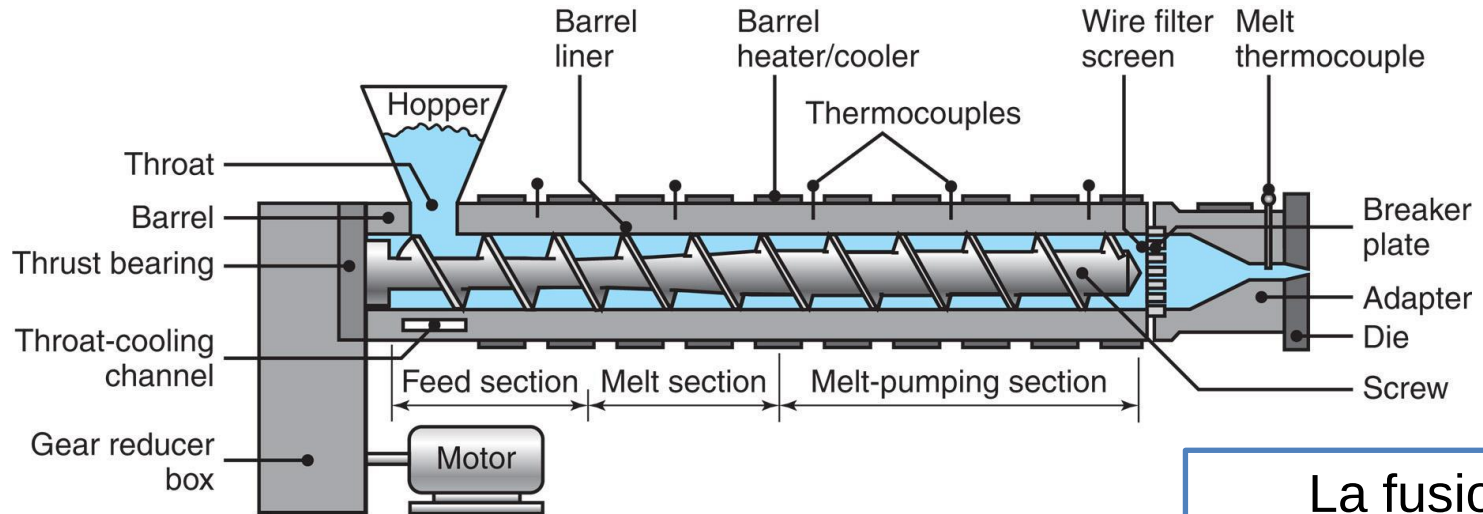
Esempi: termoplastici e termoindurenti

Introduzione

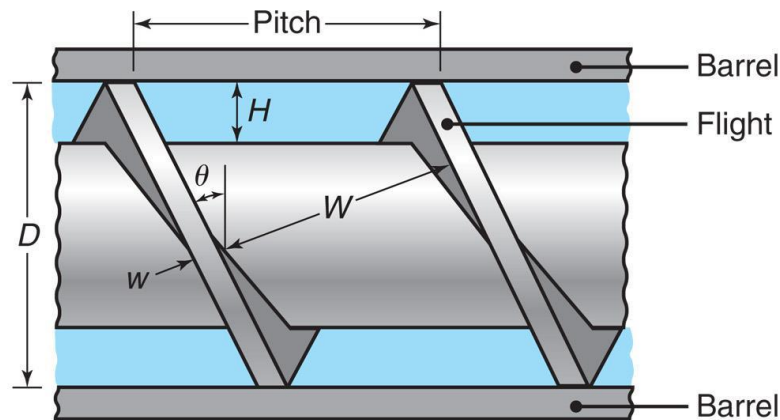
[Video: panoramica trasformazione termoplastici](#)



Estrusione



(a)

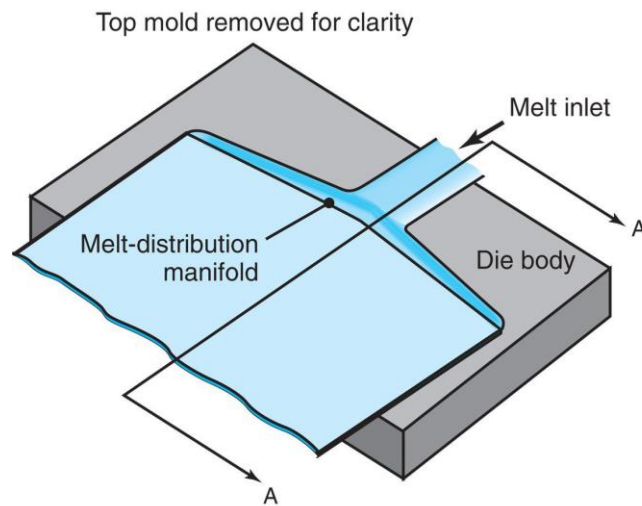
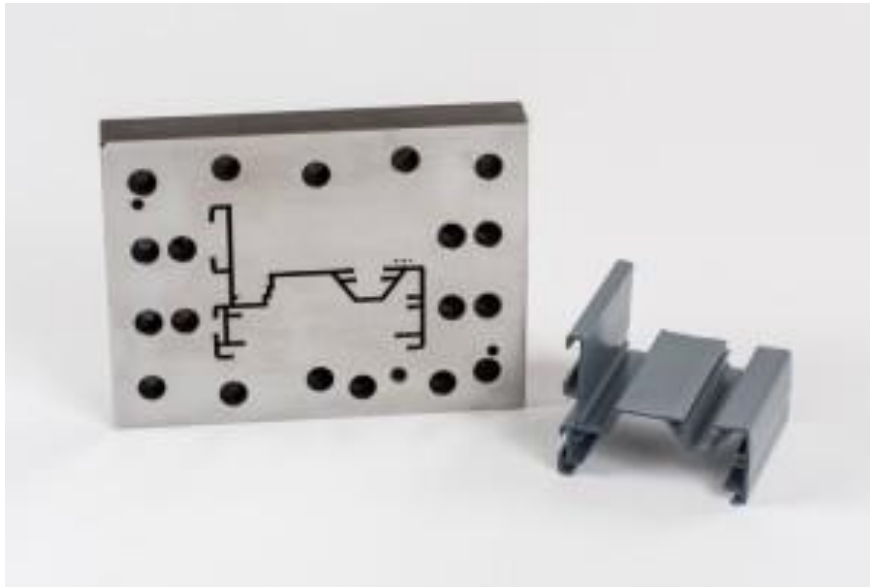


(b)

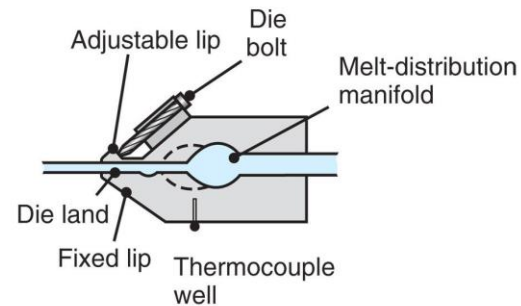
La fusione
mediante **vite di
estrusione** è alla
base di molti
processi per la
trasformazione di
**polimeri
termoplastici.**

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Estrusione

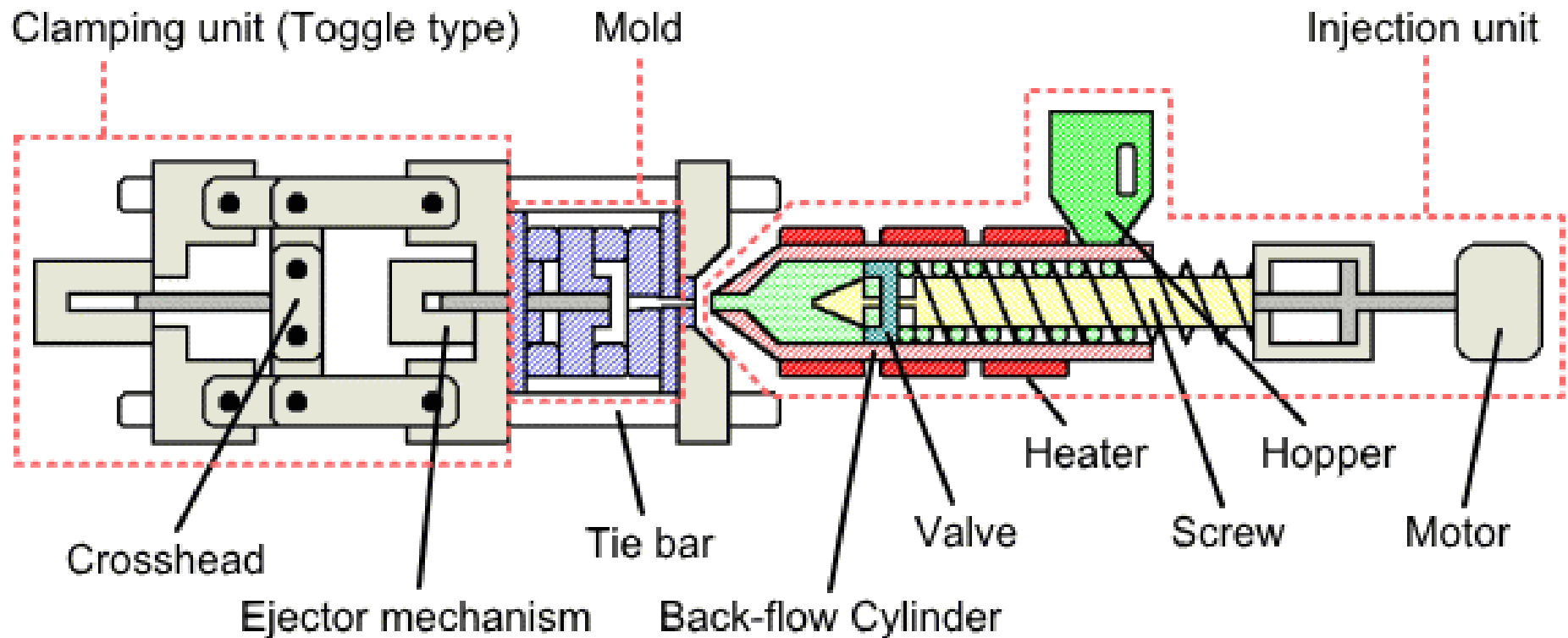


View AA:



[Video:
estruzione](#)

Injection Molding

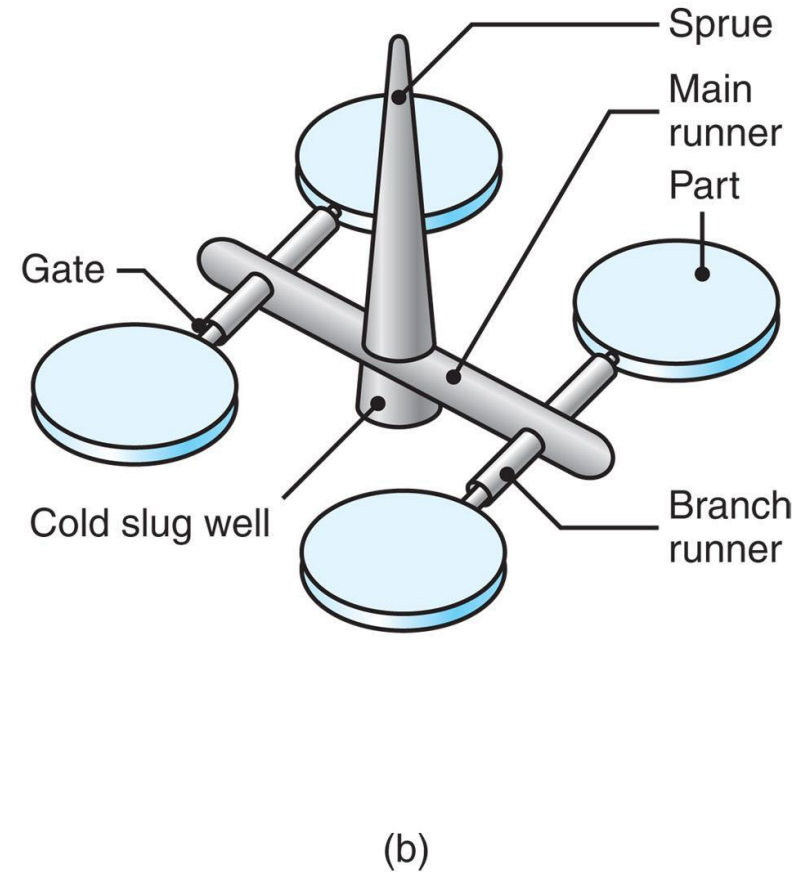
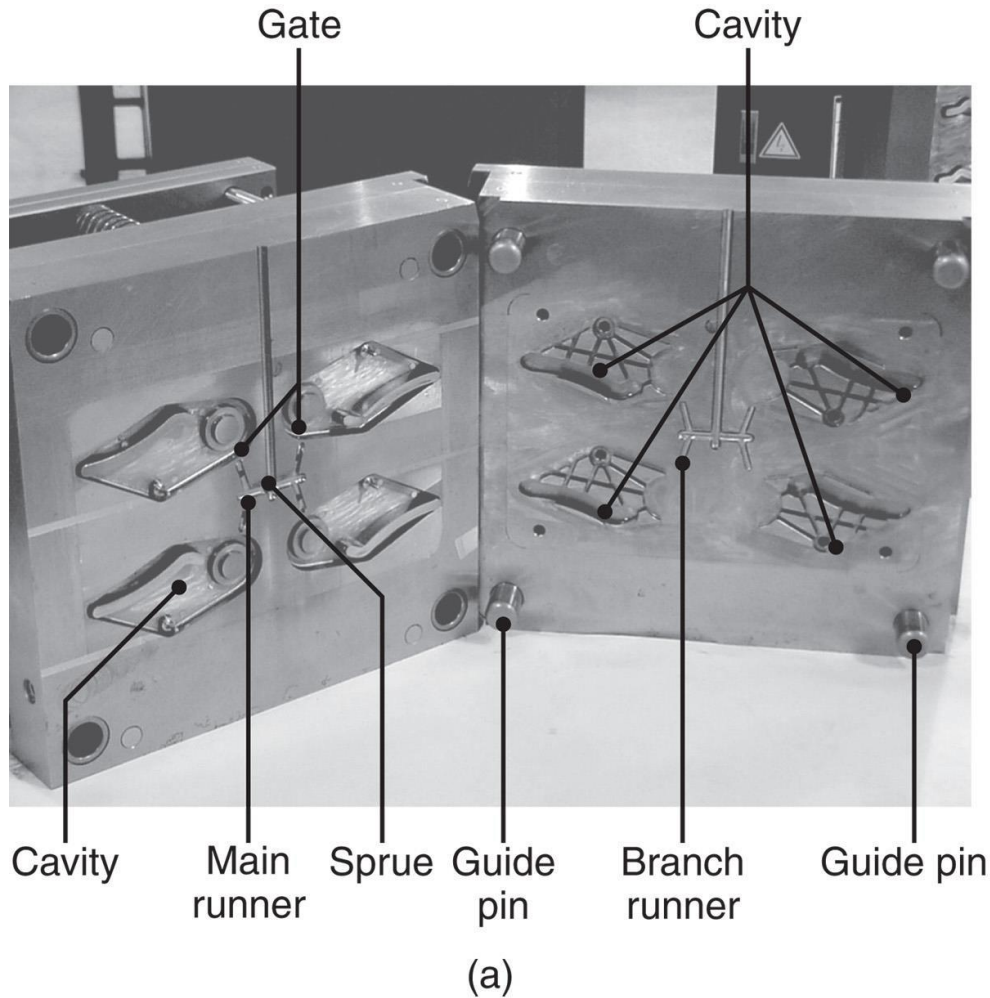


Injection Molding

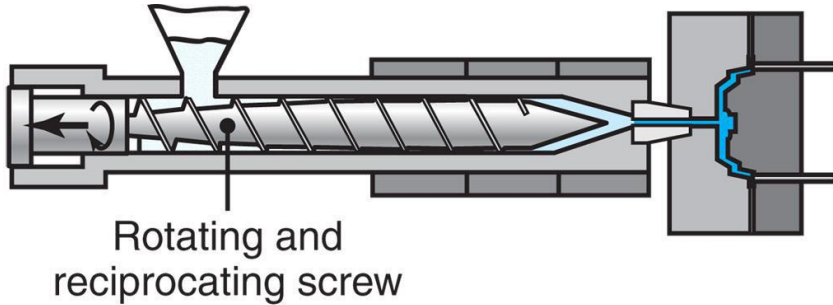


[Source](#)

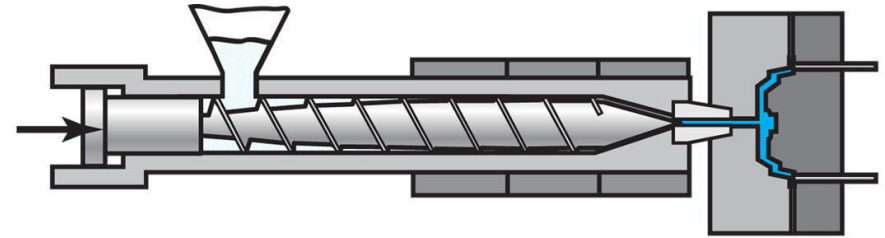
Injection Molding



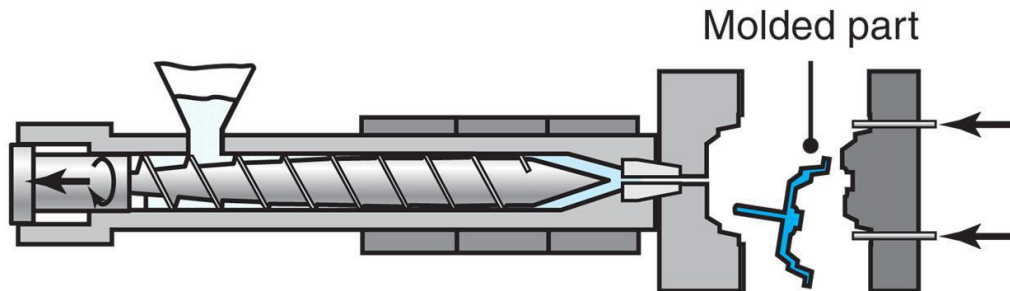
Injection Molding



1. Build up polymer in front of sprue bushing; pressure pushes the screw backward. When sufficient polymer has built up, rotation stops.



2. When the mold is ready, the screw is pushed forward by a hydraulic cylinder, filling the sprue bushing, sprue, and mold cavity with polymer. The screw begins rotating again to build up more polymer.



3. After polymer is solidified/cured, the mold opens, and ejector pins remove the molded part.

[Video: processo](#)

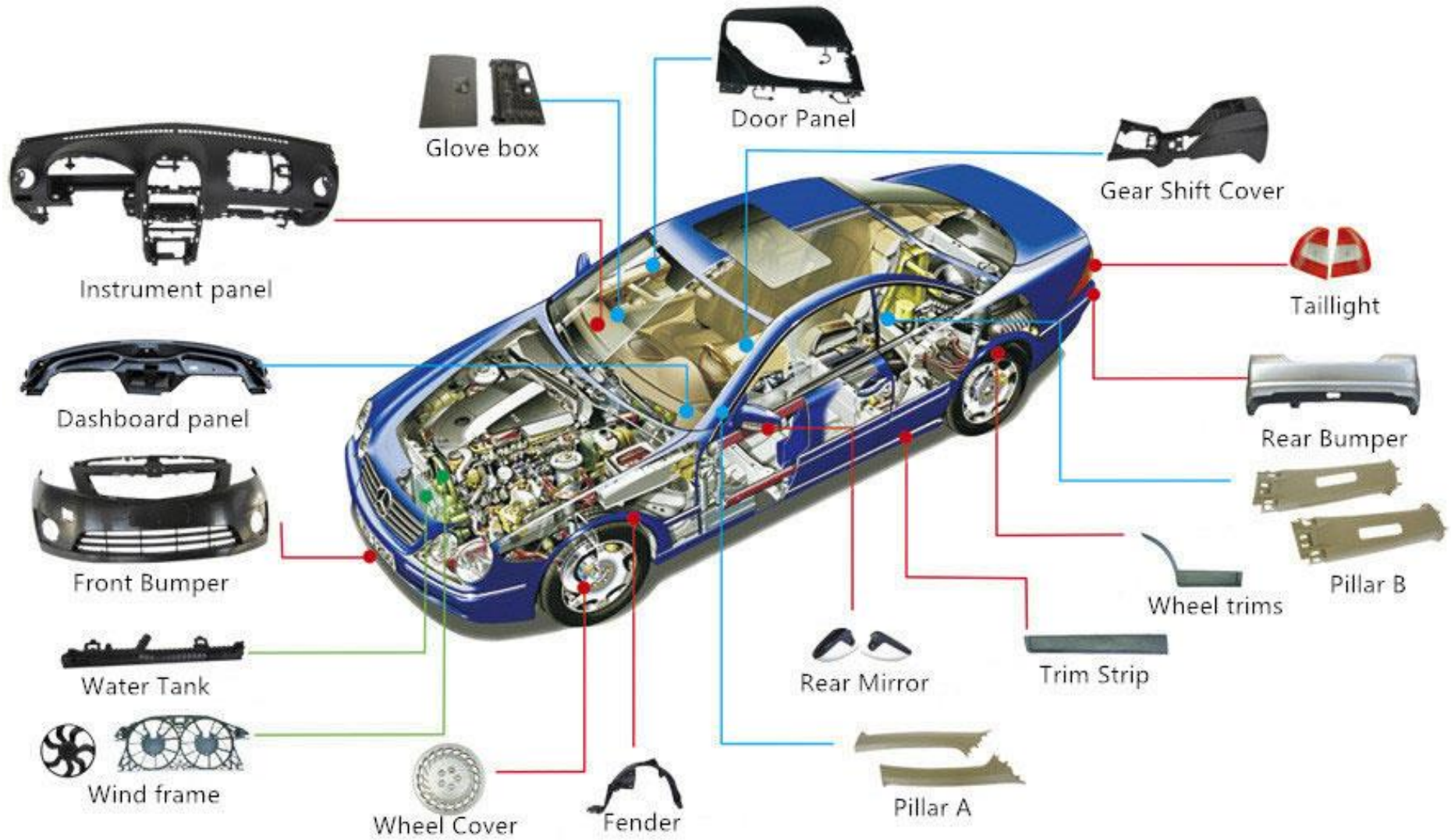
[Video: cups](#)

[Video: tappi](#)

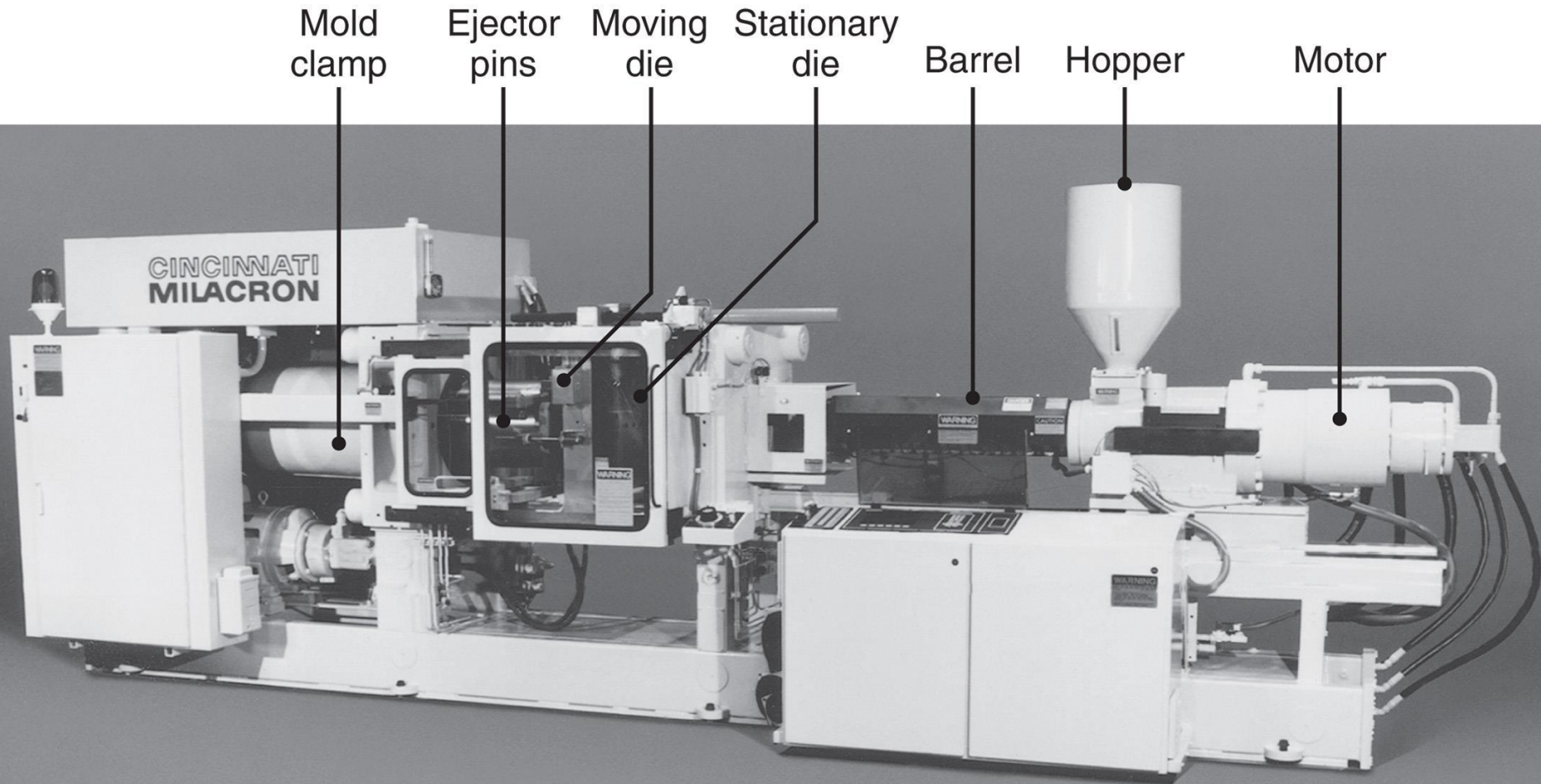
Injection Molding



Injection Molding

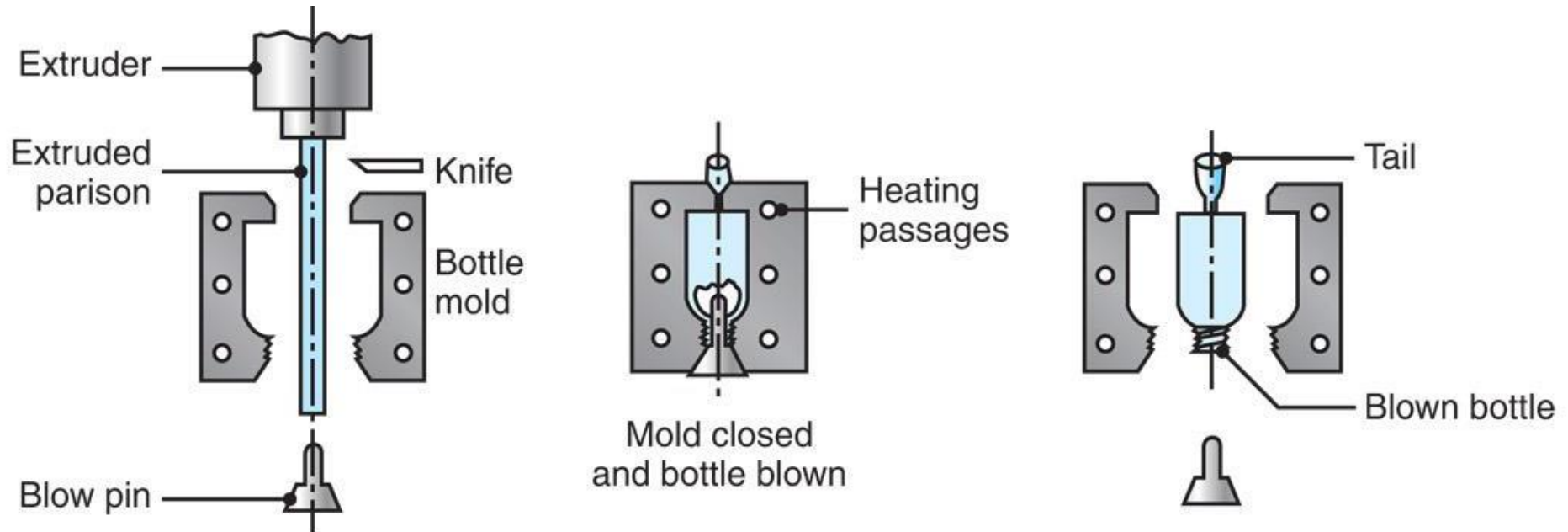


Injection Molding



Blow Molding

Extrusion Blow Molding

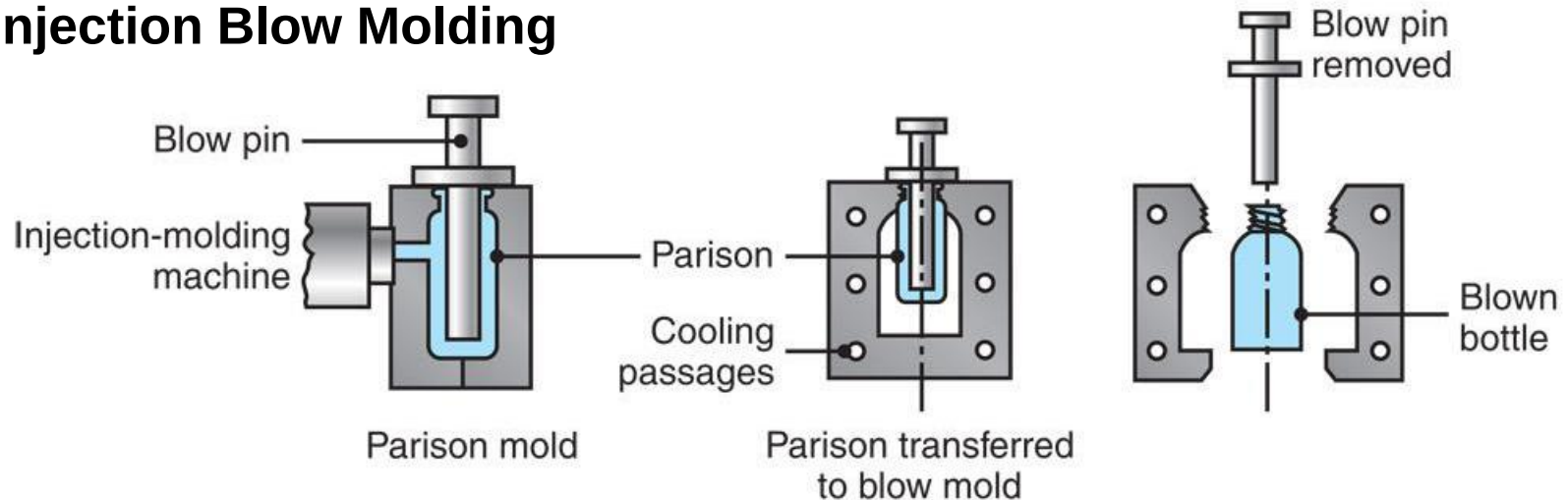


[Video: processo](#)

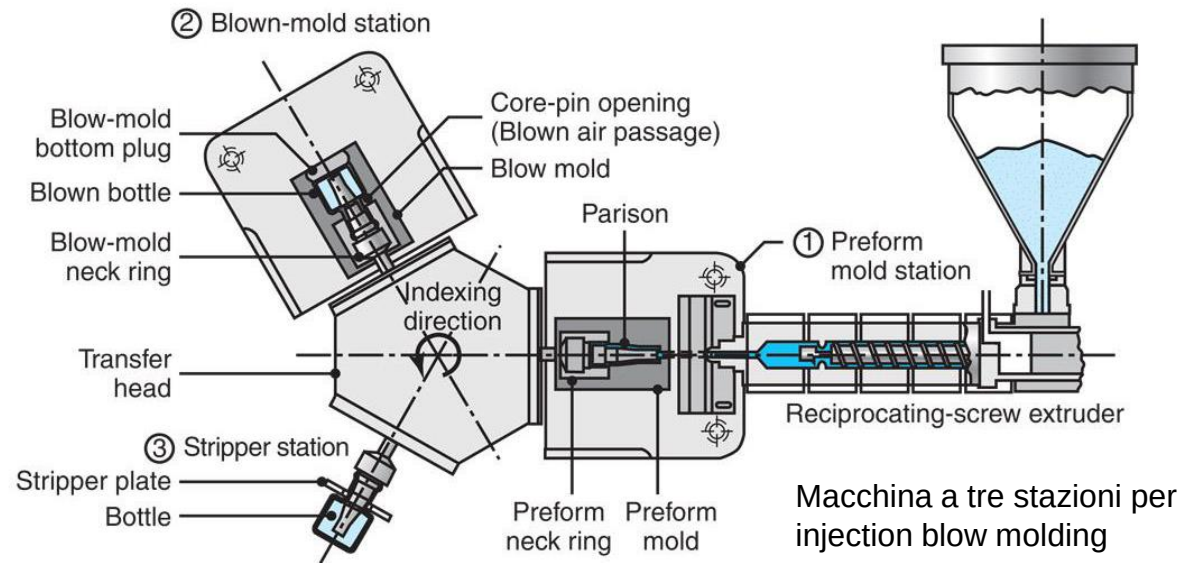
[Video: big part](#)

Blow Molding

Injection Blow Molding



[Video: processo a 3 stazioni](#)

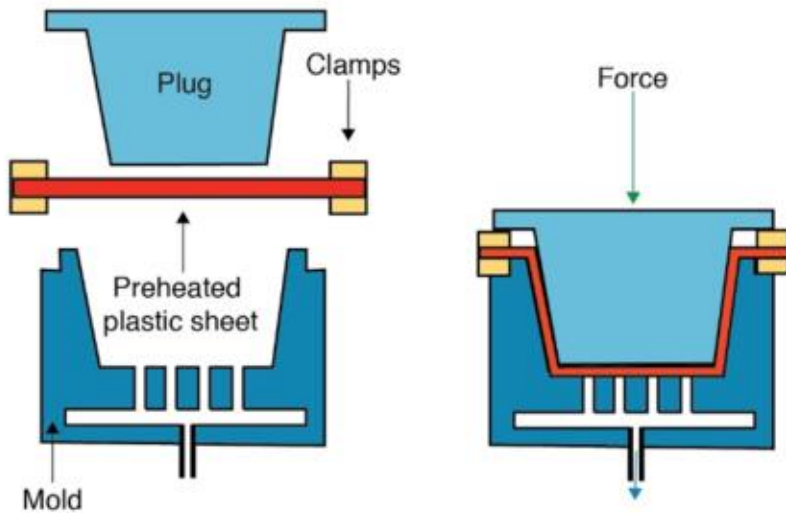


Blow Molding

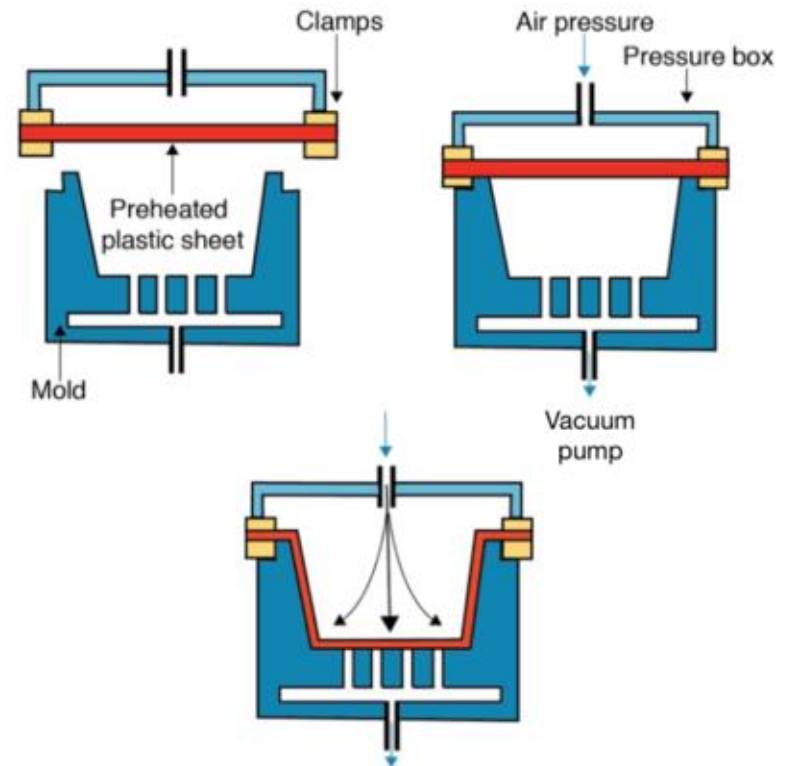


Thermoforming

Termoformatura meccanica



Termoformatura a pressione



[Video: processo](#)

[Video: impianto e processo](#)

Thermoforming



Processi per materiali compositi



Processi per materiali compositi

Matrice



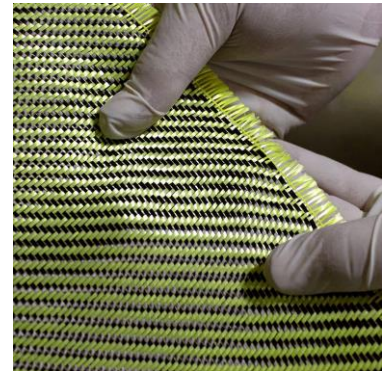
Carica

Resine termoidurenti:

- Poliesteri insaturi
- Esterni vinilici
- Epossidiche
- Fenoliche
- Esteri cianati

Fibre:

- Vetro
- Carbonio
- Aramidiche
- Boro
- PE ad alto modulo



Processi principali:

- Spray/hand layup
- Vacuum bag molding
- Compression molding
- Resin transfer molding
- Filament winding
- Sheet lamination
- Pultrusion

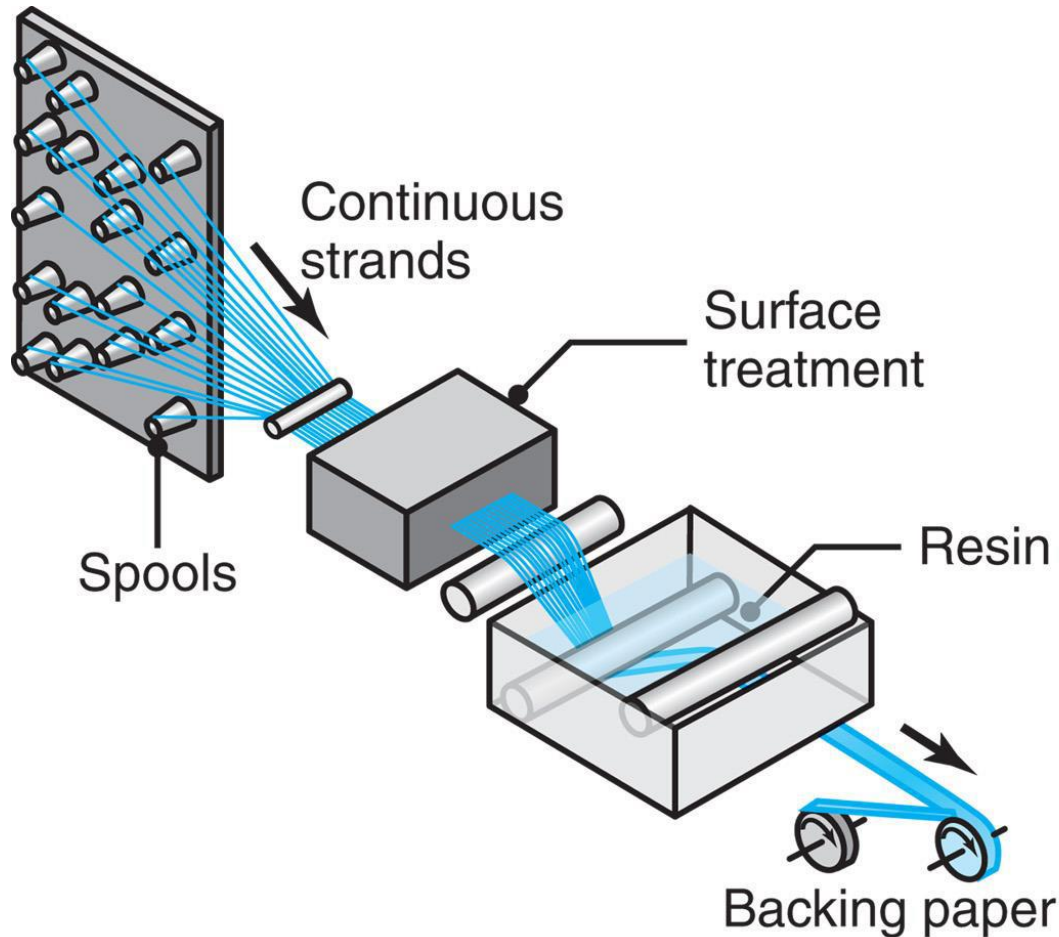
[Video: overview](#)

[Video: BMW carbon fibre production](#)

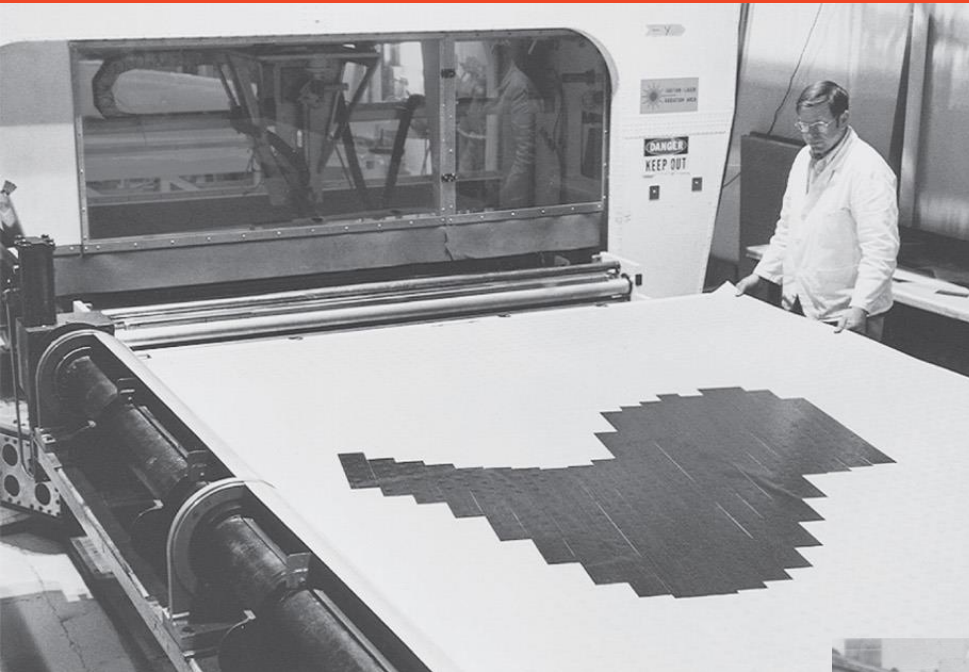


Processi per materiali compositi

Produzione del nastro/foglio pre-impregnato (prepreg)

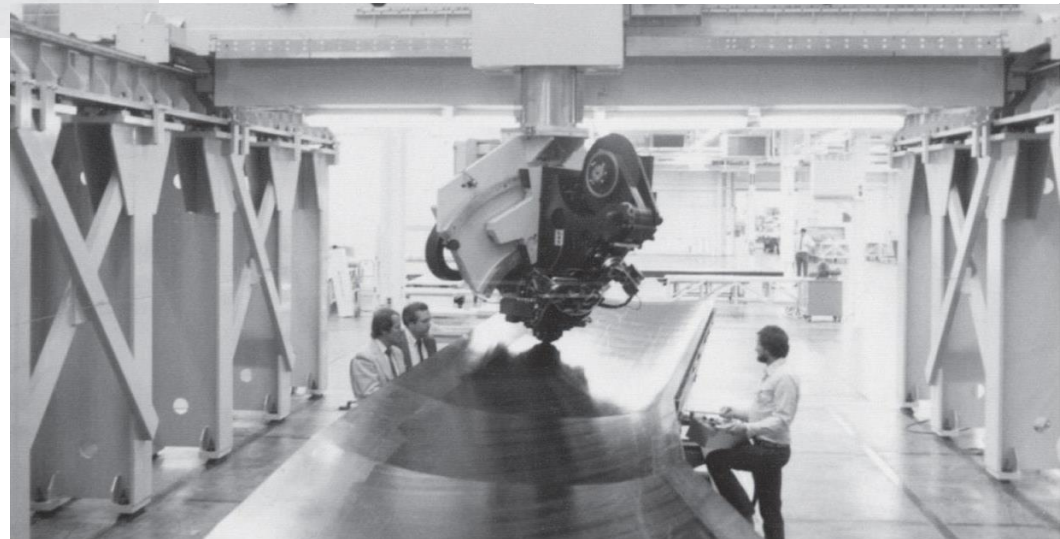


Processi per materiali compositi



Processo di layup

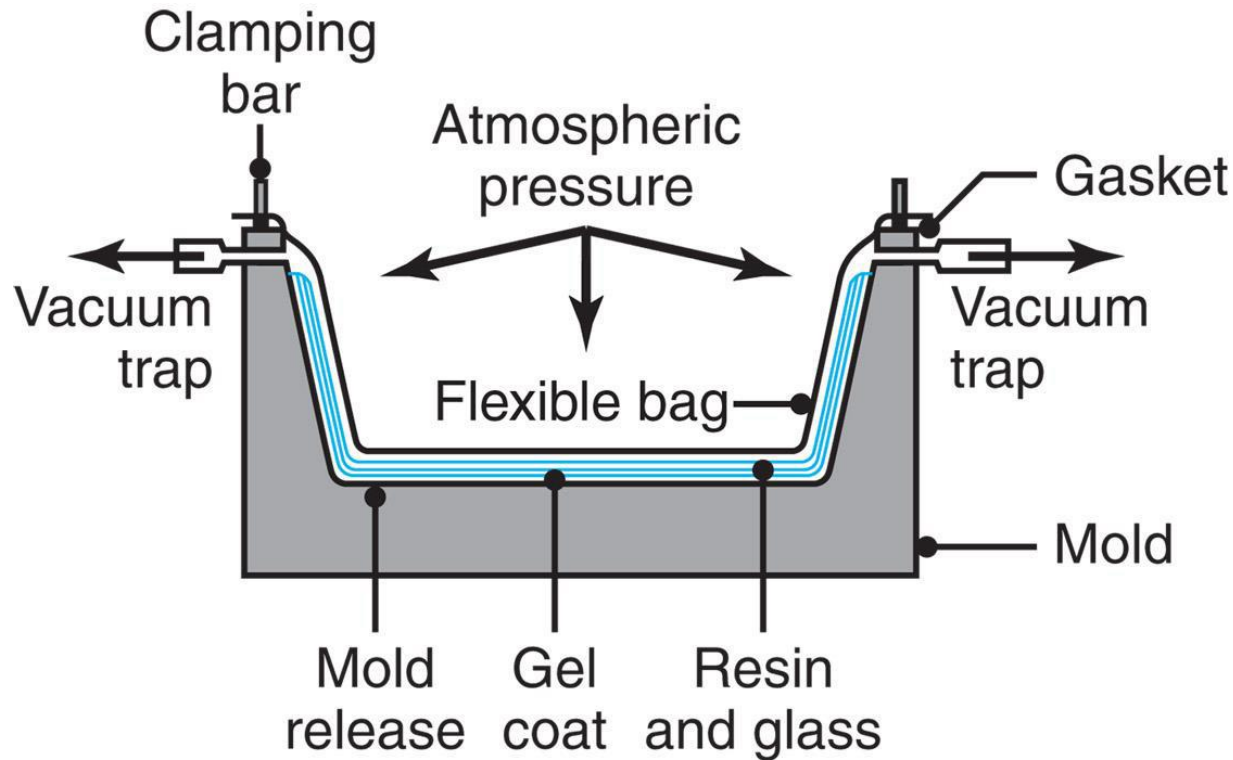
Automatizzato



Manuale

Processi per materiali compositi

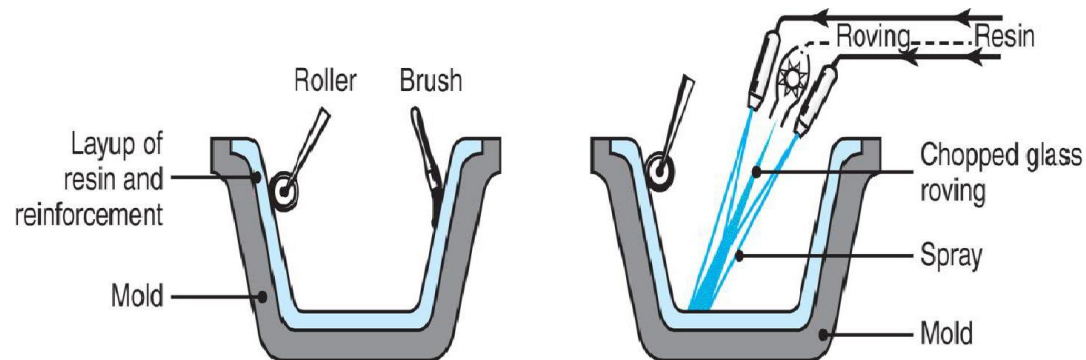
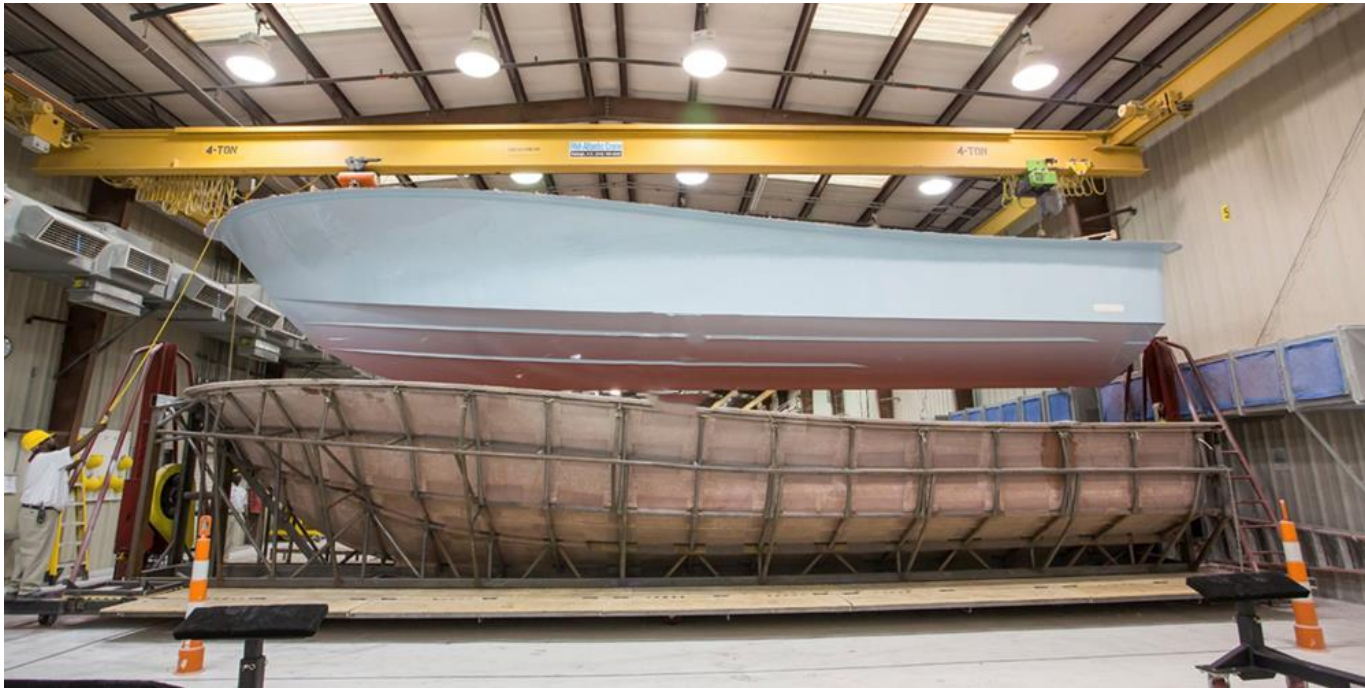
Vacuum bag molding



[Video: telaio Alfa Romeo 4C](#)

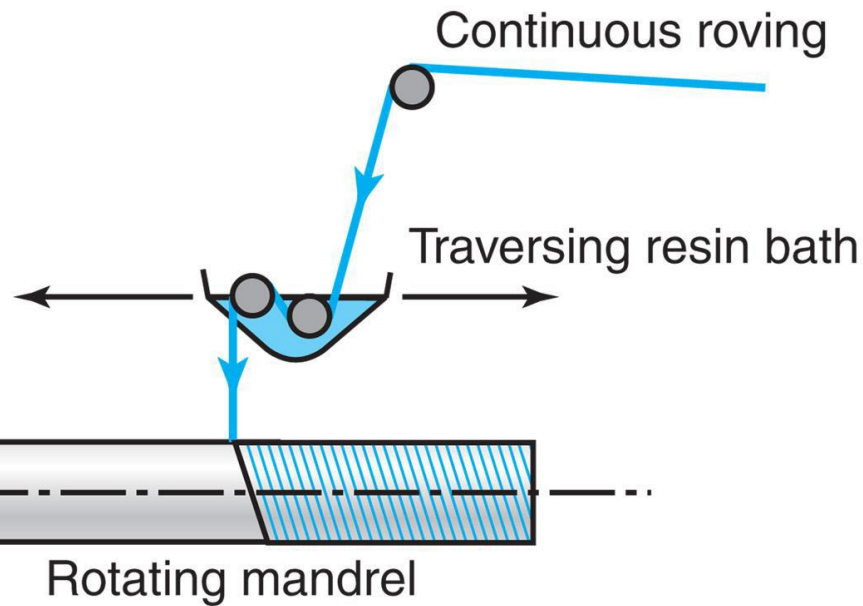
Processi per materiali compositi

Spray layup

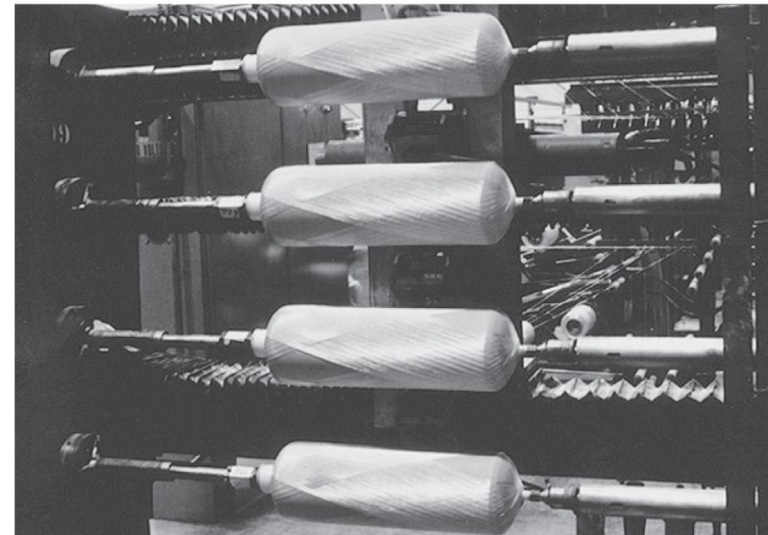


Processi per materiali compositi

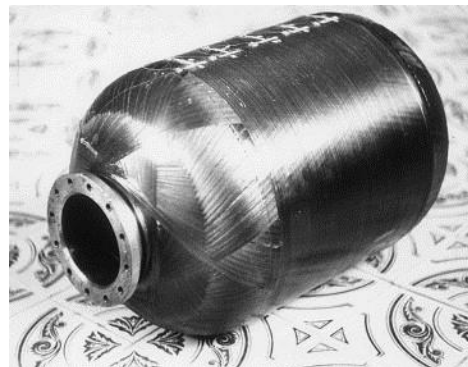
Filament winding



(a)



(b)



[Video: filament winding](#)

Processi per materiali compositi



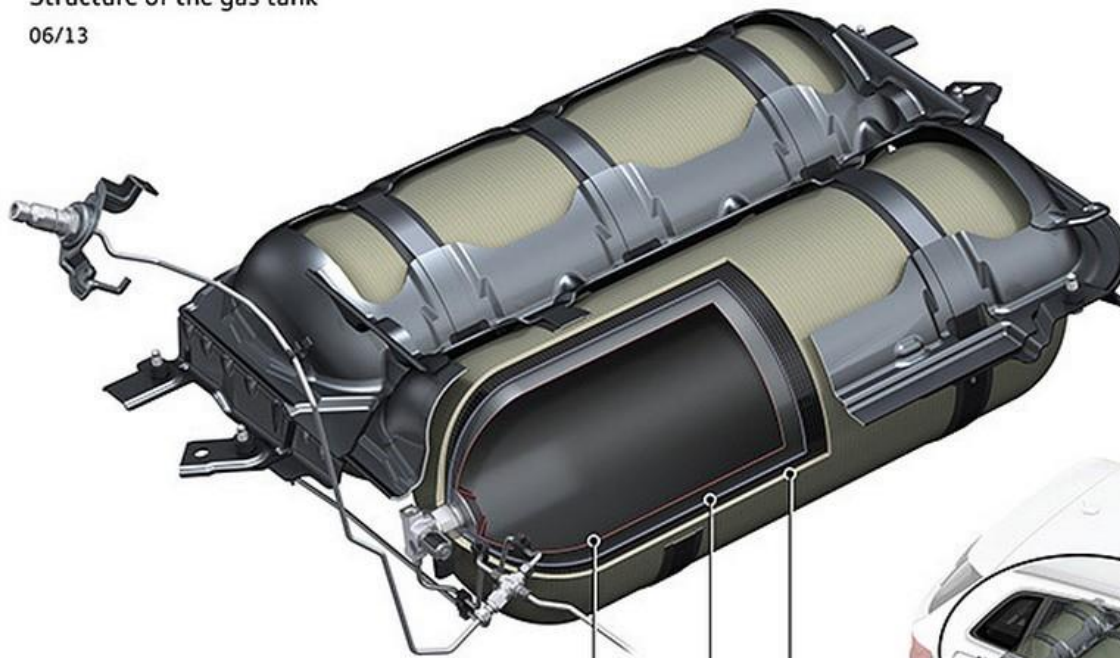
Audi

Audi A3 Sportback g-tron

Aufbau des Gastanks

Structure of the gas tank

06/13



innere Schicht: gasdichter PA-Kunststoff
Inner layer: gas sealed PA plastic

mittlere Schicht: CFK
Middle layer: CFRP

äußere Schicht: GFK
Outer layer: GFRP



Processi per polimeri e compositi

Comparative Production Characteristics of Various Molding Methods

Molding method	Equipment and tooling cost*	Production rate*	Economical production quantity*
Extrusion	M-L	VH-H	VH
Injection molding	VH	VH	VH
Rotational molding	M	M-L	M
Blow molding	M	H-M	H
Compression molding	H-M	M	H-M
Transfer molding	H	M	VH
Thermoforming	M-L	M-L	H-M
Casting	M-L	M-L	L
Centrifugal casting	H-M	M-L	M-L
Pultrusion	H-M	H	H
Filament winding	H-M	L	L
Spray layup and hand layup	L-VL	L-VL	L

* VH = very high; H = high; M = medium; L = low; VL = very low.

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